Lecture 15: Time Stepping: Simpson's Rule and Runge–Kutta

18.085 Computational Science and Engineering I • MW 9:30–11, 6-120 • Wednesday, November 4, 2026

After this lecture you should be able to

- Derive Simpson's rule in two ways: by cancelling the h^3 errors of the midpoint rule and the trapezoid, and by integrating a parabola.
- Find the error of Simpson's rule, $-\frac{h^5}{2880}f^{(4)}$ per step, and see why the rule has no error for cubics.
- Build RK4 from Simpson's rule by predicting the missing values with Euler steps.
- Show that RK4 is fourth order, first on the test equation $\dot{y} = \mu y$ and then for any $\dot{y} = f(y)$.
- Carry out one RK4 step by hand.
- Compare the five methods of Lectures 14 and 15 on accuracy, stability and cost.

1 What we take from Lecture 14

Every method of Lecture 14 starts from the integral form of $\dot{\boldsymbol{y}} = \boldsymbol{f}(t, \boldsymbol{y})$ over one step,

$$\boldsymbol{y}(t_{n+1}) = \boldsymbol{y}(t_n) + \int_{t_n}^{t_{n+1}} \boldsymbol{f}(t, \boldsymbol{y}(t))\, dt,$$

and replaces the integral by an area. Along the true solution the integrand is a function of t alone, $f(t) = \dot{y}(t)$, and we write dots for its time derivatives: $\dot{f} = \ddot{y}$, $\ddot{f} = \dddot{y}$. The area sketches of Lecture 14 showed what each rule misses in one step:

rule	area	exact minus rule	used by
left rectangle	$h\, f(t_n)$	$+\frac{h^2}{2}\,\dot{f}$	forward Euler
right rectangle	$h\, f(t_{n+1})$	$-\frac{h^2}{2}\,\dot{f}$	backward Euler
midpoint	$h\, f(t_{n+1/2})$	$+\frac{h^3}{24}\,\ddot{f}$	leapfrog
trapezoid	$\frac{h}{2}(f(t_n) + f(t_{n+1}))$	$-\frac{h^3}{12}\,\ddot{f}$	trapezoidal

A plus sign means the rule gives too little, a minus sign too much. None of these is better than second order. Two tools from Lecture 14 take us further today: Taylor series about the middle of the step, where the odd powers cancel (as in leapfrog's centred differences), and predicting an unknown value with a forward Euler step. The first gives Simpson's rule, a fourth-order rule for the integral (Section 2). The second turns Simpson's rule into a time stepper, RK4 (Section 3).

2 Simpson's rule

In this section $f = f(t)$ is a function of t alone, and we write $f_n = f(t_n)$, $f_m = f(t_{n+1/2})$ and $f_{n+1} = f(t_{n+1})$ for its values at the start, the midpoint and the end of the step.

One expansion for every rule

Measure time from the midpoint, $s = t - t_{n+1/2}$, so that the step is $-h/2 \le s \le h/2$. Taylor's theorem about the midpoint, with every derivative taken at $t_{n+1/2}$, gives

$$f(t_{n+1/2}+s) = f_m + s\,\dot f + \frac{s^2}{2}\,\ddot f + \frac{s^3}{6}\,\dddot f + \frac{s^4}{24}\,f^{(4)} + \cdots .$$

Integrate term by term over $-h/2 \le s \le h/2$. The odd powers s and s^3 integrate to zero over the symmetric interval, and for even k

$$\int_{-h/2}^{h/2} s^k\,ds = 2\int_0^{h/2} s^k\,ds = \frac{2}{k+1}\Big(\frac{h}{2}\Big)^{k+1}, \qquad \text{so} \qquad \int_{-h/2}^{h/2} s^2\,ds = \frac{h^3}{12}, \qquad \int_{-h/2}^{h/2} s^4\,ds = \frac{h^5}{80}.$$

The exact integral over one step is therefore

$$I = \int_{t_n}^{t_{n+1}} f(t)\,dt = hf_m + \frac{1}{2}\cdot\frac{h^3}{12}\,\ddot f + \frac{1}{24}\cdot\frac{h^5}{80}\,f^{(4)} + \cdots = hf_m + \frac{h^3}{24}\,\ddot f + \frac{h^5}{1920}\,f^{(4)} + \cdots .$$

The midpoint rule and the trapezoid

The **midpoint rule** $M = hf_m$ is the first term, so

$$I - M = \frac{h^3}{24}\,\ddot f + \frac{h^5}{1920}\,f^{(4)} + \cdots .$$

For the **trapezoid** $T = \frac{h}{2}(f_n + f_{n+1})$ we need the end values. They are the expansion at $s = -h/2$ and $s = +h/2$:

$$f_n = f_m - \frac{h}{2}\,\dot f + \frac{h^2}{8}\,\ddot f - \frac{h^3}{48}\,\dddot f + \frac{h^4}{384}\,f^{(4)} - \cdots ,$$

$$f_{n+1} = f_m + \frac{h}{2}\,\dot f + \frac{h^2}{8}\,\ddot f + \frac{h^3}{48}\,\dddot f + \frac{h^4}{384}\,f^{(4)} + \cdots .$$

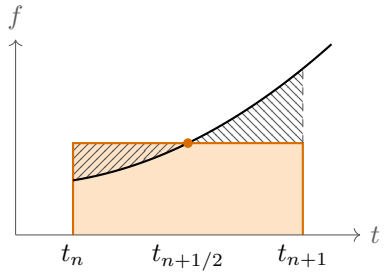
Add them: the odd powers cancel, as in leapfrog's centred second difference, and the even powers double,

$$f_n + f_{n+1} = 2f_m + \frac{h^2}{4}\,\ddot f + \frac{h^4}{192}\,f^{(4)} + \cdots , \qquad T = \frac{h}{2}\,(f_n + f_{n+1}) = hf_m + \frac{h^3}{8}\,\ddot f + \frac{h^5}{384}\,f^{(4)} + \cdots .$$

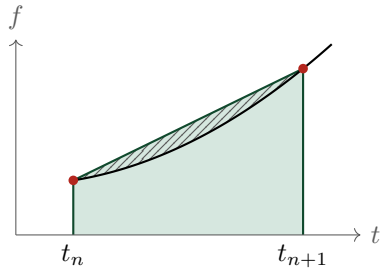
Subtract from I:

$$I - T = \Big(\frac{1}{24} - \frac{1}{8}\Big)h^3\,\ddot f + \Big(\frac{1}{1920} - \frac{1}{384}\Big)h^5 f^{(4)} + \cdots = -\frac{h^3}{12}\,\ddot f - \frac{h^5}{480}\,f^{(4)} + \cdots .$$

These are the last two rows of the table in Section 1, now with their next terms.



midpoint rule $M = hf_m$
misses about $\frac{h^3}{24}\ddot f$



trapezoid $T = \frac{h}{2}(f_n + f_{n+1})$
counts about $\frac{h^3}{12}\ddot f$ too much

The two second-order rules on the same curve. The midpoint rectangle misses a little net area (its two hatched pieces nearly cancel); the trapezoid counts the hatched sliver above the curve, twice as much and on the other side.

Cancelling the h^3 terms

The two h^3 errors have opposite signs and the trapezoid's is twice as large: $\frac{1}{24}$ against $-\frac{1}{12}$. So take the midpoint rule twice and the trapezoid once,

$$S = \frac{2M + T}{3}.$$

The weights add to 1, so S is still a rule for the integral, and its error is the same combination of the two errors:

$$\begin{aligned} I - S &= \frac{2\,(I - M) + (I - T)}{3} = \frac{1}{3}\Big[\Big(\frac{2}{24} - \frac{1}{12}\Big)h^3\,\ddot{f} + \Big(\frac{2}{1920} - \frac{1}{480}\Big)h^5 f^{(4)}\Big] + \cdots \\ &= \frac{1}{3}\Big[0 - \frac{2}{1920}\,h^5 f^{(4)}\Big] + \cdots = -\frac{h^5}{2880}\,f^{(4)} + \cdots . \end{aligned}$$

The h^3 term is gone. Written with the three values of f,

$$S = \frac{1}{3}\Big(2hf_m + \frac{h}{2}\,(f_n + f_{n+1})\Big) = \frac{h}{6}(f_n + 4f_m + f_{n+1}).$$

Only even derivatives of f appear in these errors, because the odd powers of s integrate to zero. So the third derivative never enters, and S has no error at all when f is a cubic.

Simpson's rule

$$\int_{t_n}^{t_{n+1}} f(t)\,dt \approx \frac{h}{6}(f_n + 4f_m + f_{n+1}), \qquad \text{error } -\frac{h^5}{2880}\,f^{(4)} \text{ per step.}$$

It is the combination of the midpoint rule and the trapezoid in which their h^3 errors cancel.

The same weights from a parabola

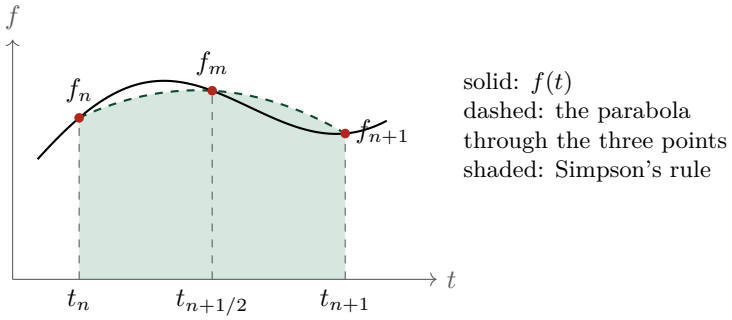
A second route gives the same rule. Replace f by the parabola through the three points (t_n, f_n), $(t_{n+1/2}, f_m)$ and (t_{n+1}, f_{n+1}), and integrate the parabola. With s as before, write the parabola as $p(s) = a + bs + cs^2$. The three conditions are

$$p(0) = a = f_m, \qquad p(-\tfrac{h}{2}) = a - \frac{bh}{2} + \frac{ch^2}{4} = f_n, \qquad p(\tfrac{h}{2}) = a + \frac{bh}{2} + \frac{ch^2}{4} = f_{n+1}.$$

Adding the last two gives $f_n + f_{n+1} = 2a + ch^2/2$, so $c = 2\,(f_n + f_{n+1} - 2f_m)/h^2$. Now integrate. The odd term bs integrates to zero over the symmetric interval, and $\int_{-h/2}^{h/2} s^2\,ds = h^3/12$:

$$\int_{-h/2}^{h/2} p(s)\,ds = a\,h + c\,\frac{h^3}{12} = hf_m + \frac{h}{6}(f_n + f_{n+1} - 2f_m) = \frac{h}{6}(f_n + 4f_m + f_{n+1}).$$

The weights are $\frac{1}{6}, \frac{4}{6}, \frac{1}{6}$. The rule integrates cubics without error as well: for $f = s^3$ the integral and the rule are both zero. The first function it gets wrong is $f = s^4$, whose integral is $\int_{-h/2}^{h/2} s^4\,ds = h^5/80$ while the rule gives $\frac{h}{6}(\frac{h^4}{16} + 0 + \frac{h^4}{16}) = h^5/48$. The error (exact minus rule) is $h^5/80 - h^5/48 = -h^5/120$, and since $f^{(4)} = 24$ for this f, this agrees with the error $-\frac{h^5}{2880}f^{(4)}$ found by cancelling.



Example 2.1 (Three rules on one integral)**. Given.** $\int_0^{0.5} e^t\,dt = e^{0.5} - 1 = 0.6487213$, taken as one step with $h = 0.5$.

Find. What do the midpoint rule, the trapezoid and Simpson's rule give? How do their errors compare with the predictions $\frac{h^3}{24}\ddot{f}$, $-\frac{h^3}{12}\ddot{f}$ and $-\frac{h^5}{2880}f^{(4)}$?

Answer. Every derivative of e^t is e^t, which is $e^{0.25} = 1.2840254$ at the midpoint. The three rules give

$$M = 0.5\,e^{0.25} = 0.6420127, \qquad T = 0.25\,(1 + e^{0.5}) = 0.6621803, \qquad S = \frac{2M + T}{3} = 0.6487352,$$

and S is also $\frac{0.5}{6}(1 + 4e^{0.25} + e^{0.5})$. Against the exact value:

rule	exact minus rule	first error term
midpoint	0.0067086	$\frac{h^3}{24}e^{0.25} = 0.0066876$
trapezoid	−0.0134590	$-\frac{h^3}{12}e^{0.25} = -0.0133753$
Simpson	−0.0000140	$-\frac{h^5}{2880}e^{0.25} = -0.0000139$

The midpoint error is half the size of the trapezoid's and of the other sign, as predicted. Its next term $\frac{h^5}{1920}e^{0.25} = 0.0000209$ accounts for most of the small difference. Simpson's rule is about 480 times more accurate than the midpoint rule here, from the same three values of f that the midpoint rule and the trapezoid use together.

Many steps

Over an interval of length L split into N steps of $h = L/N$, the step errors $-\frac{h^5}{2880}f^{(4)}$ add up to about $-\frac{h^4}{2880}\int f^{(4)}\,dt$: Simpson's rule is **fourth order**, and halving h divides the error by about 16. For $\int_0^1 e^t\,dt = e - 1$:

steps N	exact minus Simpson	ratio
1	-5.79×10^{-4}	
2	-3.70×10^{-5}	15.7
4	-2.33×10^{-6}	15.9
8	-1.46×10^{-7}	16.0

For $N = 8$ the estimate $-\frac{h^4}{2880}(\dddot{f}(1) - \dddot{f}(0)) = -\frac{e-1}{2880 \cdot 4096} = -1.46 \times 10^{-7}$ is right to the digits shown. If $\dot{y} = f(t)$ depends on t alone, the step $y_{n+1} = y_n + \frac{h}{6}(f_n + 4f_m + f_{n+1})$ is a fourth-order time stepper. When f depends on y as well, the values f_m and f_{n+1} are not available, and that is the next section.

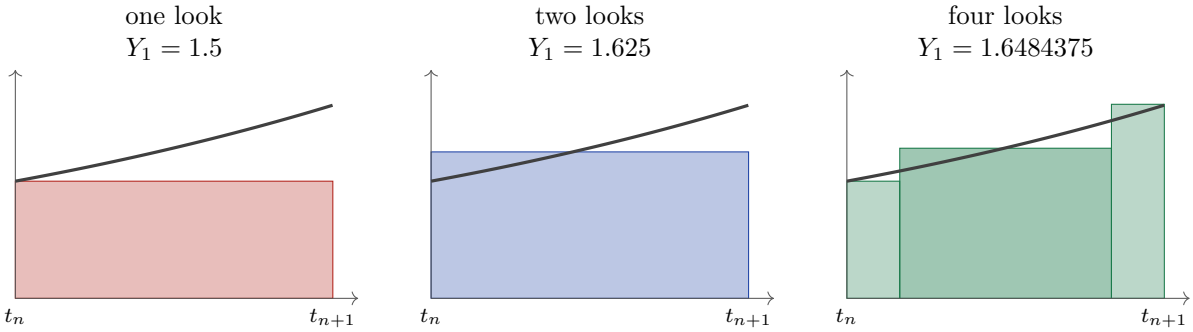
3 Runge–Kutta: the idea

Every method so far is a way to estimate one area. Over a step of size h,

$$\boldsymbol{Y}_{n+1} = \boldsymbol{Y}_n + \int_{t_n}^{t_{n+1}} \boldsymbol{f}\, dt,$$

so the step is decided by the height of $\boldsymbol{f}$ across the interval. Forward Euler looks at the slope once, at the start, and commits to it for the whole step. If the slope is changing, that one look is out of date before the step ends.

Runge–Kutta methods take several looks at the slope inside the step, then move once with a weighted average of what they saw. Nothing is solved, and earlier steps are not needed. The picture below shows the area each choice covers for $\dot{y} = y$, $y(0) = 1$, $h = 0.5$, drawn to scale. The true value is $e^{0.5} = 1.6487213$.

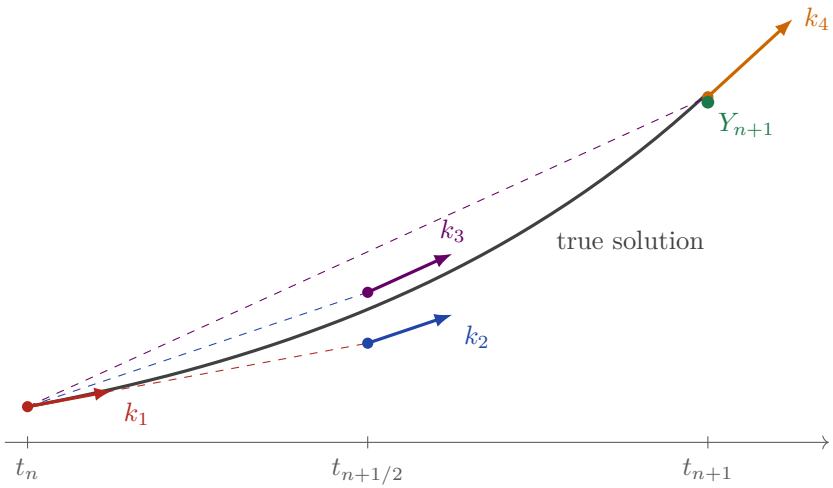


With one look we get a single rectangle, which misses the growth of the curve. With a look at the middle the rectangle sits at about the right average height. With four looks the area becomes three strips of widths $h/6$, $4h/6$ and $h/6$: these are the weights $1, 4, 1$ of Simpson's rule, and the strips cover the area almost exactly.

The four looks

Simpson's rule wants the height at the start, the middle and the end. The middle and the end lie in the future, so we do not know $\boldsymbol{Y}$ there. RK4 scouts ahead, each time from $\boldsymbol{Y}_n$, using the latest slope it has:

Look 1:	slope where we stand	$\boldsymbol{k}_1 = \boldsymbol{f}(t_n,\ \boldsymbol{Y}_n)$
Look 2:	walk half a step with $\boldsymbol{k}_1$, look again	$\boldsymbol{k}_2 = \boldsymbol{f}(t_n + \frac{h}{2},\ \boldsymbol{Y}_n + \frac{h}{2}\boldsymbol{k}_1)$
Look 3:	walk half a step with $\boldsymbol{k}_2$ instead, look again	$\boldsymbol{k}_3 = \boldsymbol{f}(t_n + \frac{h}{2},\ \boldsymbol{Y}_n + \frac{h}{2}\boldsymbol{k}_2)$
Look 4:	walk a full step with $\boldsymbol{k}_3$, look at the end	$\boldsymbol{k}_4 = \boldsymbol{f}(t_n + h,\ \boldsymbol{Y}_n + h\boldsymbol{k}_3)$



This is $\dot{y} = y$ again with a long step, $h = 1.6$, so that the looks are easy to tell apart. Each dashed walk has the colour of the slope it uses, and the arrow at its end is the next look. None of the trial points is kept. Only the four slopes are, and the green dot is where the real step lands.

The RK4 step

$$\boldsymbol{Y}_{n+1} = \boldsymbol{Y}_n + \frac{h}{6}\left(\boldsymbol{k}_1 + 2\boldsymbol{k}_2 + 2\boldsymbol{k}_3 + \boldsymbol{k}_4\right)$$

These are Simpson's weights $1, 4, 1$, with the 4 shared between the two looks at the middle. Each step costs four evaluations of $\boldsymbol{f}$. The error over a fixed time is of order h^4: halve h and it falls by about 16.

$\dot{y} = y$, $h = 0.5$	Looks per step	Y_1	Error
Forward Euler	1	1.5	0.149
Midpoint (two stages)	2	1.625	0.024
RK4	4	1.6484375	0.00028

The derivation, with the proof that the method is fourth order, is in the Supplementary Information at the end of these notes (Section S).

4 Comparing the five methods

With RK4 there are five methods: forward Euler, backward Euler, leapfrog and the trapezoidal method from Lecture 14, and RK4. This section compares them on the oscillator $\ddot{u} + u = 0$ and on a fast decay.

4.1 RK4 on the oscillator

For $\ddot{u} + u = 0$ the growth matrix of Section S is $\boldsymbol{G}_{RK} = \boldsymbol{I} + h\boldsymbol{A} + \frac{1}{2}(h\boldsymbol{A})^2 + \frac{1}{6}(h\boldsymbol{A})^3 + \frac{1}{24}(h\boldsymbol{A})^4$. Since $\boldsymbol{A}^2 = -\boldsymbol{I}$, the powers of $h\boldsymbol{A}$ are $(h\boldsymbol{A})^2 = -h^2\boldsymbol{I}$, $(h\boldsymbol{A})^3 = -h^3\boldsymbol{A}$ and $(h\boldsymbol{A})^4 = h^4\boldsymbol{I}$. Collecting the multiples of $\boldsymbol{I}$ and of $\boldsymbol{A}$,

$$\boldsymbol{G}_{RK} = \Big(1 - \frac{h^2}{2} + \frac{h^4}{24}\Big)\boldsymbol{I} + \Big(h - \frac{h^3}{6}\Big)\boldsymbol{A} = \begin{bmatrix} a & b \\ -b & a \end{bmatrix}, \qquad a = 1 - \frac{h^2}{2} + \frac{h^4}{24}, \quad b = h - \frac{h^3}{6}.$$

This is a multiple of a rotation, with eigenvalues $a \pm ib$. Expanding the squares,

$$a^2 = 1 - h^2 + \frac{h^4}{3} - \frac{h^6}{24} + \frac{h^8}{576}, \qquad b^2 = h^2 - \frac{h^4}{3} + \frac{h^6}{36},$$

so the terms in h^2 and h^4 cancel and

$$|\lambda|^2 = a^2 + b^2 = 1 - \frac{h^6}{72} + \frac{h^8}{576}.$$

Then $|\lambda| \le 1$ if and only if $h^8/576 \le h^6/72$, that is $h^2 \le 8$. So $|\lambda| < 1$ for $0 < h < 2\sqrt{2} \approx 2.83$, and the method blows up for $h > 2\sqrt{2}$. For small h the damping is slight, about $1 - h^6/144$ per step (0.99989 at $h = 0.5$). For the angle, $a + ib$ is the series $1 + ih + (ih)^2/2 + (ih)^3/6 + (ih)^4/24$, which differs from the exponential by $(ih)^5/120$:

$$a + ib = e^{ih} - \frac{(ih)^5}{120} + \cdots = e^{ih}\Big(1 - \frac{ih^5}{120} + \cdots\Big), \qquad \text{so} \qquad \theta = \arctan\frac{b}{a} = h - \frac{h^5}{120} + \cdots .$$

The computed oscillation falls behind by about $h^5/120$ per step.

Example 4.1 (One RK4 step by hand)**. Given.** $\boldsymbol{f}(t, \boldsymbol{y}) = \boldsymbol{A}\boldsymbol{y} = (v, -u)$, $\boldsymbol{Y}_0 = (1, 0)$ and $h = 0.5$.

Find. What are the four slopes $\boldsymbol{k}_1, \ldots, \boldsymbol{k}_4$ and $\boldsymbol{Y}_1$? How close is $\boldsymbol{Y}_1$ to the true point $(\cos 0.5, -\sin 0.5)$?

Answer. With h as a letter the answer is the first column of $\boldsymbol{G}_{RK}$, $\boldsymbol{Y}_1 = (a, -b)$ with $a = 1 - h^2/2 + h^4/24$ and $b = h - h^3/6$. Step by step at $h = 0.5$, each slope is $\boldsymbol{f}$ at a trial point:

$$\begin{aligned}
&\boldsymbol{k}_1 = \boldsymbol{A}\boldsymbol{Y}_0 = (0, -1), \\
&\boldsymbol{Y}_0 + 0.25\,\boldsymbol{k}_1 = (1, -0.25), && \boldsymbol{k}_2 = (-0.25, -1), \\
&\boldsymbol{Y}_0 + 0.25\,\boldsymbol{k}_2 = (0.9375, -0.25), && \boldsymbol{k}_3 = (-0.25, -0.9375), \\
&\boldsymbol{Y}_0 + 0.5\,\boldsymbol{k}_3 = (0.875, -0.46875), && \boldsymbol{k}_4 = (-0.46875, -0.875).
\end{aligned}$$

Then $\boldsymbol{k}_1 + 2\boldsymbol{k}_2 + 2\boldsymbol{k}_3 + \boldsymbol{k}_4 = (-1.46875, -5.75)$ and

$$\boldsymbol{Y}_1 = (1, 0) + \tfrac{1}{12}\,(-1.46875, -5.75) = (0.877604, -0.479167).$$

The same point comes from $\boldsymbol{G}_{RK}$ with $a = 0.877604$ and $b = 0.479167$. The true point is $(0.877583, -0.479426)$, so after one step RK4 is off by 0.00026. After one step of the same size forward Euler is off by 0.124 and the trapezoidal method by 0.010.

4.2 The same two steps

Examples 1.1, 2.1, 3.2 and 4.1 of Lecture 14 and Example 4.1 here start from $(u, v) = (1, 0)$ with $h = 0.5$. After two steps, at $t = 1$:

method	U_2	V_2	$U_2^2 + V_2^2$	off by
forward Euler	0.750	−1.000	1.5625	0.2629
backward Euler	0.480	−0.640	0.6400	0.2103
leapfrog	0.531	−0.953	1.1907	0.0100
trapezoidal	0.557	−0.830	1.0000	0.0201
RK4	0.541	−0.841	0.9996	0.0005
true solution	0.540	−0.841	1	

Leapfrog's V_2 is compared with the velocity at $t = 1.25$, so its $U_2^2 + V_2^2$ mixes two times.

4.3 One eigenvalue at a time

For the table below and for the stability regions, look at one eigenvalue at a time, with the test equation of Section S. For the scalar equation $\dot{y} = \mu y$ each method multiplies Y_n by a factor $R(z)$, with $z = h\mu$, against e^z for the true solution: forward Euler gives $Y_{n+1} = (1 + h\mu)\, Y_n$, so $R(z) = 1 + z$, and in the same way

$$\text{backward Euler:} \quad R(z) = \frac{1}{1-z},$$
$$\text{trapezoidal:} \quad R(z) = \frac{1 + z/2}{1 - z/2},$$
$$\text{RK4:} \quad R(z) = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24}.$$

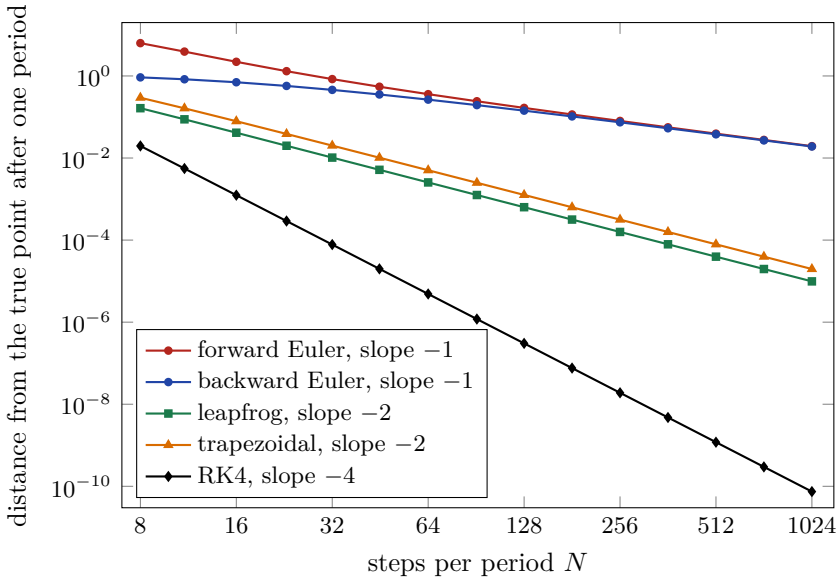
For a system $\dot{\boldsymbol{y}} = \boldsymbol{A}\boldsymbol{y}$ these methods have $\boldsymbol{G} = R(h\boldsymbol{A})$, so the eigenvalues of $\boldsymbol{G}$ are the numbers $R(h\mu)$, one for each eigenvalue μ of $\boldsymbol{A}$ (the letter λ stays with the eigenvalues of $\boldsymbol{G}$). Leapfrog is built for second-order equations and has no single factor $R(z)$. The oscillator has $\mu = \pm i$, so $z = \pm ih$ lies on the imaginary axis; a mode that decays like e^{-at} has $z = -ah$ on the negative real axis. The **stability region** of a method is the set of z with $|R(z)| \le 1$.

For forward Euler it is the disc $|1 + z| \le 1$, of radius 1 about $z = -1$; the imaginary axis touches it only at $z = 0$, so forward Euler is unstable for the oscillator at every $h > 0$. For backward Euler it is $|1 - z| \ge 1$, everything outside the disc of radius 1 about $z = +1$. That contains the whole left half-plane, so every decaying mode stays bounded at every step size; methods with this property are called **A-stable**. For the trapezoidal method $|R(z)| \le 1$ means $|1 + z/2| \le |1 - z/2|$, that is $|z + 2| \le |z - 2|$: z is no farther from -2 than from $+2$, which holds if and only if $\operatorname{Re} z \le 0$. So its stability region is the left half-plane, and its edge, the imaginary axis, is where the oscillator's $z = \pm ih$ lie.

4.4 A table of properties

	forward Euler	backward Euler	leapfrog	trapezoidal	RK4
explicit or implicit	explicit	implicit	explicit	implicit	explicit
work per step	1 evaluation	1 solve	1 evaluation	1 solve	4 evaluations
order of accuracy	1	1	2	2	4
$R(z)$	$1 + z$	$\dfrac{1}{1-z}$	none	$\dfrac{1 + z/2}{1 - z/2}$	$\displaystyle\sum_{k=0}^{4} \frac{z^k}{k!}$
$\lvert\lambda\rvert$ for $\ddot{u} + u = 0$	$\sqrt{1 + h^2}$	$\dfrac{1}{\sqrt{1 + h^2}}$	1 ($h < 2$)	1	$\sqrt{1 - \frac{h^6}{72} + \frac{h^8}{576}}$
energy over many periods	grows	decays	bounded	conserved	decays slowly
angle per step (true: h)	$h - \frac{h^3}{3}$	$h - \frac{h^3}{3}$	$h + \frac{h^3}{24}$	$h - \frac{h^3}{12}$	$h - \frac{h^5}{120}$
stable for $\ddot{u} + \omega^2 u = 0$	never	every h	$\omega h < 2$	every h	$\omega h \le 2\sqrt{2}$
stable for $\dot{y} = -ay$	$ah \le 2$	every h	does not apply	every h	$ah \le 2.785$

The angles are the first terms of their series. Backward Euler is "stable" for the oscillator only in the sense that it does not blow up; it still damps the oscillation away.



Distance of the computed point from the true point after one period, for N steps of $h = 2\pi/N$ from $(u, v) = (1, 0)$. Leapfrog starts with half a kick and its V is compared with the velocity half a step later.

4.5 Accuracy for the work

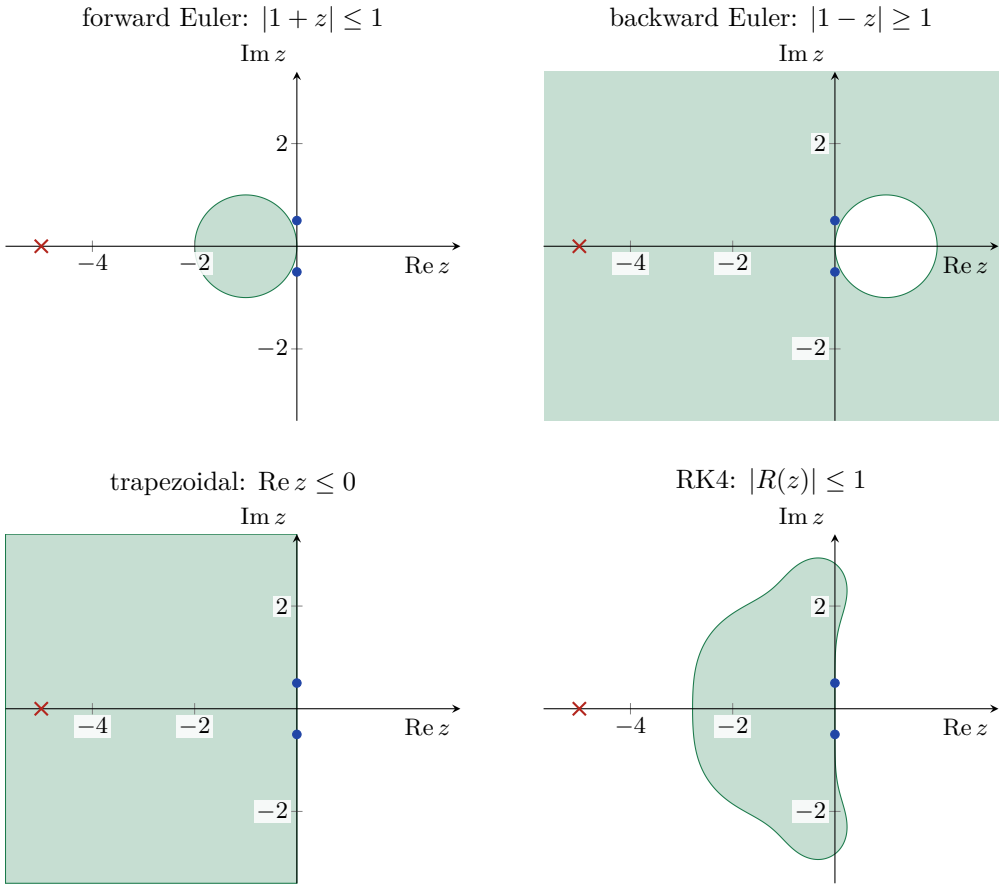
How many steps does each method need? Ask for the computed point (U, V) to be within 1% of the radius of the true point at the end of ten periods (for leapfrog, V is compared with the velocity half a step later). The fewest steps per period that do it are:

method	steps per period	evaluations or solves per period
forward Euler	19 838	19 838
backward Euler	19 641	19 641
leapfrog	102	102
trapezoidal	144	144
RK4	17	68

The first-order methods need about twenty thousand steps per period, the second-order methods about a hundred, RK4 seventeen. The orders of accuracy show as slopes on a log-log plot of the error against the number of steps: doubling the steps halves the Euler errors, quarters the leapfrog and trapezoidal errors, and divides the RK4 error by 16. At the same step leapfrog's error is half that of the trapezoidal method, because its phase error per step is $h^3/24$ against $h^3/12$.

4.6 Stability regions

The stability regions $|R(z)| \le 1$ of Section 4.3 are shaded below for four of the methods, with the oscillator's $z = \pm 0.5i$ (dots, $h = 0.5$) and the fast decay $z = -5$ of Example 4.2 (cross). Leapfrog has no region of this kind: for $\ddot{u} = -\omega^2 u$ its condition is the stretch $|\omega h| < 2$ of the imaginary axis. (Some books give the name leapfrog to the two-step method $Y_{n+1} = Y_{n-1} + 2h\, f(Y_n)$ for a first-order equation; that method is stable only on the stretch $|z| < 1$ of the imaginary axis. It is not the staggered method of this lecture.)



The oscillator's dots lie outside the forward Euler disc, inside the backward Euler region (where they are damped), on the edge of the trapezoidal half-plane, and just inside the RK4 region ($|R(\pm 0.5i)| = 0.99989$). The RK4 region reaches $\pm 2\sqrt{2}\, i$ on the imaginary axis and -2.785 on the real axis. Only the implicit methods contain the cross.

Example 4.2 (A fast decay)**. Given.** An RC circuit with time constant $RC = 20$ ms, so that its capacitor voltage x obeys $\dot{x} = -x/(RC) = -50\, x$, is one part of a larger simulation whose slow parts are stepped with $h = 0.1$ s.

Find. What does each method multiply the voltage x by at each step? Which methods keep it bounded?

Answer. Here $z = -50 \times 0.1 = -5$, and the true factor per step is $e^{-5} = 0.0067$.

method	factor per step $R(-5)$	after five steps
forward Euler	-4	-1024
backward Euler	$1/6 = 0.167$	0.00013
trapezoidal	$-3/7 = -0.429$	-0.014
RK4	13.7	4.8×10^{5}
true solution	0.0067	1.4×10^{-11}

Leapfrog does not apply to this first-order equation. Backward Euler gives the right picture: the fast part is gone after a step or two. The trapezoidal method keeps it bounded but flips its sign at every step. Forward Euler and RK4 blow up; they would need $h \le 2/50 = 0.04$ s and

$h \le 2.785/50 = 0.056$ s, steps set by a part of the circuit that does nothing of interest after its first hundred milliseconds or so (a few time constants). Problems like this, with fast modes that decay and slow modes that we want to follow, are called **stiff**, and implicit methods are made for them.

Which method when

- *Long runs of oscillations, where the energy must not drift* (orbits, molecular dynamics, waves): leapfrog, which is explicit and cheap, inside its step limit; or the trapezoidal method, which keeps the energy of a linear oscillator at any step, at the cost of a solve.
- *Smooth problems without fast decaying modes*, where accuracy for the work matters: RK4, or an adaptive Runge–Kutta pair such as `ode45`.
- *Stiff problems*: backward Euler, or the backward differentiation formulas that extend it to higher order, which damp the fast modes at any step.
- *Forward Euler*: for understanding, and as a building block; rarely on its own.

Exercise 4.3 (An LC circuit)**. Given.** A circuit with $L = 1$ mH and $C = 10\ \mu$F and no resistance, so that the charge obeys $L\ddot{q} + q/C = 0$, that is $\ddot{q} + \omega^2 q = 0$ with $\omega = 1/\sqrt{LC}$.

Find. (a) What are ω and the period T? (b) For which steps h is each of the five methods stable? (c) Using the table of steps per period, what step does each method need to stay within 1% for ten periods? Which sets the step here, stability or accuracy?

Where this is going

That closes our excursion through discrete mechanical models: masses and springs, their motion, and how to step that motion in time. Next time we return to differential equations in space and to the *finite element method*, which discretises the weak form of $-\frac{d}{dx}(c(x)\frac{du}{dx}) = f(x)$ rather than the equation itself. The stiffness matrix $\boldsymbol{K}$ will reappear, but assembled from integrals of basis functions instead of read off a spring chain.

Reading

Strang, *Computational Science and Engineering*, § 2.2 (Oscillation by Newton's Law) for leapfrog and the trapezoidal method, and § 6.2 (Finite Difference Methods for ODEs) for Runge–Kutta methods and stability. For why leapfrog keeps the energy close over long times: E. Hairer, C. Lubich and G. Wanner, Geometric numerical integration illustrated by the Störmer–Verlet method, *Acta Numerica* 12 (2003), 399–450.

Answer to Exercise 4.3

(a) $\omega = 1/\sqrt{10^{-3} \cdot 10^{-5}} = 10^4$ rad/s, so $f = 1.59$ kHz and $T = 2\pi/\omega = 628\ \mu$s.

(b) In the time ωt the equation is our oscillator, with the step ωh. Leapfrog needs $\omega h < 2$, that is $h < 200\ \mu$s; RK4 needs $\omega h \le 2\sqrt{2}$, that is $h \le 283\ \mu$s. The trapezoidal method is stable at every step, and so is backward Euler, which damps the oscillation. Forward Euler is unstable at every step.

(c) The step is $h = T/N$ with N from the table: forward Euler 0.032 μs, backward Euler 0.032 μs, leapfrog 6.2 μs, trapezoidal 4.4 μs, RK4 37 μs (with 68 evaluations per period). These steps are far below the stability limits of (b): here accuracy, not stability, sets the step.

Appendix: Simpson's rule, RK4 and the five methods in Python

The script below repeats Examples 2.1 and S.1 and the table of Simpson's rule with N steps, builds the five growth matrices, repeats the two-step comparison and Example 4.1, finds the steps per period for 1% by bisection, and prints the factors of Example 4.2. It needs only NumPy.

```
"""Lecture 15: Simpson's rule, RK4, and five time-stepping methods for u'' + u = 0 (y' = Ay, y
    = (u, v))."""
import numpy as np

# 1. three rules for the integral of e^t over one step [0, 0.5]
h = 0.5
M = h * np.exp(h / 2)                                   # midpoint
T = h / 2 * (1 + np.exp(h))                             # trapezoid
S = (2 * M + T) / 3                                     # Simpson = (h/6)(f_n + 4 f_m + f_n+1)
exact = np.exp(h) - 1
print(f"integral of e^t on [0, 0.5]: exact {exact:.7f}")
for name, r in (("midpoint", M), ("trapezoid", T), ("Simpson", S)):
    print(f"  {name:9s} {r:.7f}  exact minus rule {exact - r:+.7f}")

# 2. Simpson's rule with N steps on [0, 1]
for N in (1, 2, 4, 8):
    hN = 1 / N
    t = hN * np.arange(N)
    SN = np.sum(hN / 6 * (np.exp(t) + 4 * np.exp(t + hN / 2) + np.exp(t + hN)))
    print(f"  Simpson, N = {N}: exact minus rule {np.e - 1 - SN:.3e}")

A = np.array([[0.0, 1.0], [-1.0, 0.0]])
I = np.eye(2)
methods = ["forward Euler", "backward Euler", "leapfrog", "trapezoidal", "RK4"]

def growth_matrix(method, h):
    if method == "forward Euler":
        return I + h * A
    if method == "backward Euler":
        return np.linalg.inv(I - h * A)    # a real code solves with I - hA instead
    if method == "leapfrog":
        return np.array([[1.0, h], [-h, 1.0 - h * h]])
    if method == "trapezoidal":
        return np.linalg.solve(I - h * A / 2, I + h * A / 2)
    if method == "RK4":
        hA = h * A
        return I + hA + hA @ hA / 2 + hA @ hA @ hA / 6 + hA @ hA @ hA @ hA / 24

def rk4_step(f, t, y, h):                       # one RK4 step for any y' = f(t, y)
    k1 = f(t, y)
    k2 = f(t + h / 2, y + h / 2 * k1)
    k3 = f(t + h / 2, y + h / 2 * k2)
    k4 = f(t + h, y + h * k3)
    return y + h / 6 * (k1 + 2 * k2 + 2 * k3 + k4)

def start(method, h):             # u(0) = 1, v(0) = 0; leapfrog gets half a kick
    return np.array([1.0, -h / 2 if method == "leapfrog" else 0.0])

def distance(method, y, t, h):   # leapfrog's V is the velocity half a step later
    s = h / 2 if method == "leapfrog" else 0.0
    return np.hypot(y[0] - np.cos(t), y[1] + np.sin(t + s))

```

```
# 1. two steps of h = 0.5, and the size and angle of the eigenvalues of G
h = 0.5
print("two steps of h = 0.5 from u = 1, v = 0:")
for m in methods:
    G = growth_matrix(m, h)
    y = G @ G @ start(m, h)
    lam = np.linalg.eigvals(G)[0]
    print(f"  {m:14s} U2 {y[0]:8.5f}  V2 {y[1]:8.5f}  off {distance(m, y, 1.0, h):.5f}"
          f"  |lambda| {abs(lam):.5f}  angle {abs(np.angle(lam)):.5f}")
y1 = rk4_step(lambda t, y: A @ y, 0.0, np.array([1.0, 0.0]), h)
print("  one RK4 step with stages:", np.round(y1, 6),
      " same as G:", np.allclose(y1, growth_matrix("RK4", h) @ [1.0, 0.0]))
print("RK4 for y' = y, one step of h = 0.5:", rk4_step(lambda t, y: y, 0.0, 1.0, 0.5),
      " exact", round(np.exp(0.5), 7))


# 3. fewest steps per period that keep the point within 1% after ten periods
def error_after(method, N, periods=10):
    h = 2 * np.pi / N
    n = N * periods
    y = np.linalg.matrix_power(growth_matrix(method, h), n) @ start(method, h)
    return distance(method, y, h * n, h)

print("steps per period for 1% after ten periods:")
for m in methods:
    lo, hi = 2, 10 ** 6
    while hi - lo > 1:              # bisection: the error falls as N grows
        mid = (lo + hi) // 2
        lo, hi = (lo, mid) if error_after(m, mid) <= 0.01 else (mid, hi)
    work = 4 * hi if m == "RK4" else hi
    print(f"  {m:14s} {hi:6d} steps, {work:6d} evaluations or solves per period")

# 4. a fast decay y' = -50 y with h = 0.1: z = -5
z = -5.0
R = {"forward Euler": 1 + z, "backward Euler": 1 / (1 - z),
     "trapezoidal": (1 + z / 2) / (1 - z / 2),
     "RK4": 1 + z + z ** 2 / 2 + z ** 3 / 6 + z ** 4 / 24}
print("factor per step at z = -5, exact", round(np.exp(z), 4))
for m, r in R.items():
    print(f"  {m:14s} {r:8.4f}")
```

Its output:

```
integral of e^t on [0, 0.5]: exact 0.6487213
  midpoint  0.6420127  exact minus rule +0.0067086
  trapezoid 0.6621803  exact minus rule -0.0134590
  Simpson   0.6487352  exact minus rule -0.0000140
  Simpson, N = 1: exact minus rule -5.793e-04
  Simpson, N = 2: exact minus rule -3.701e-05
  Simpson, N = 4: exact minus rule -2.326e-06
  Simpson, N = 8: exact minus rule -1.456e-07
two steps of h = 0.5 from u = 1, v = 0:
  forward Euler  U2  0.75000  V2 -1.00000  off 0.26288  |lambda| 1.11803  angle 0.46365
  backward Euler U2  0.48000  V2 -0.64000  off 0.21030  |lambda| 0.89443  angle 0.46365
  leapfrog       U2  0.53125  V2 -0.95312  off 0.00995  |lambda| 1.00000  angle 0.50536
  trapezoidal    U2  0.55709  V2 -0.83045  off 0.02009  |lambda| 1.00000  angle 0.48996
  RK4            U2  0.54059  V2 -0.84104  off 0.00052  |lambda| 0.99989  angle 0.49976
  one RK4 step with stages: [ 0.877604 -0.479167]  same as G: True
RK4 for y' = y, one step of h = 0.5: 1.6484375  exact 1.6487213
```

```
steps per period for 1% after ten periods:
  forward Euler   19838 steps,  19838 evaluations or solves per period
  backward Euler  19641 steps,  19641 evaluations or solves per period
  leapfrog          102 steps,    102 evaluations or solves per period
  trapezoidal       144 steps,    144 evaluations or solves per period
  RK4                17 steps,     68 evaluations or solves per period
factor per step at z = -5, exact 0.0067
  forward Euler   -4.0000
  backward Euler   0.1667
  trapezoidal     -0.4286
  RK4             13.7083
```

S Supplementary Information: the derivation of RK4

This section derives the RK4 step summarised in Section 3: how the method follows from Simpson's rule, and why it is fourth order.

The missing values

For $\dot{\boldsymbol{y}} = \boldsymbol{f}(t, \boldsymbol{y})$ Simpson's rule needs $\boldsymbol{f}$ at the midpoint and at the end of the step, and these need $\boldsymbol{y}(t_{n+1/2})$ and $\boldsymbol{y}(t_{n+1})$, which are not known yet. Lecture 14 had two answers to this kind of problem. Backward Euler and the trapezoidal method solve for the new value: they are implicit and cost a solve per step. Forward Euler uses only values it already has: it is explicit. Runge–Kutta methods stay explicit. They *predict* the missing values with Euler steps and use the predictions only inside $\boldsymbol{f}$.

Two stages: the midpoint method

The smallest example is the midpoint rule. Predict the midpoint with half a forward Euler step, take the slope there, and use it for the whole step:

$$\boldsymbol{k}_1 = \boldsymbol{f}(t_n, \boldsymbol{Y}_n), \qquad \boldsymbol{k}_2 = \boldsymbol{f}\big(t_n + \tfrac{h}{2},\ \boldsymbol{Y}_n + \tfrac{h}{2}\,\boldsymbol{k}_1\big), \qquad \boldsymbol{Y}_{n+1} = \boldsymbol{Y}_n + h\,\boldsymbol{k}_2.$$

How much does the prediction cost? Half a forward Euler step from $\boldsymbol{y}(t_n)$ misses $\frac{1}{2}\big(\frac{h}{2}\big)^2\ddot{\boldsymbol{y}} = \frac{h^2}{8}\ddot{\boldsymbol{y}}$ (Lecture 14), so $\boldsymbol{k}_2$ is off the true midpoint slope by a multiple of h^2, and $h\,\boldsymbol{k}_2$ by a multiple of h^3. That is the same order as the error $\frac{h^3}{24}\ddot{f}$ of the midpoint rule itself, so one step is in error by $\mathcal{O}(h^3)$ and the method is second order.

To see the order in one line, use the **test equation** $\dot{y} = \mu y$, where μ is a number, possibly complex (for the oscillator $\mu = \pm i$, the eigenvalues of $\boldsymbol{A}$). Its exact step multiplies y by

$$e^{h\mu} = e^z = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24} + \frac{z^5}{120} + \cdots, \qquad z = h\mu,$$

and each method multiplies Y_n by a factor $R(z)$ instead. For the midpoint method $k_1 = \mu Y_n$, the predicted midpoint is $Y_n + \frac{h}{2}k_1 = (1 + \frac{z}{2})Y_n$, so $k_2 = \mu(1 + \frac{z}{2})Y_n$ and

$$Y_{n+1} = Y_n + h\,k_2 = \Big(1 + z + \frac{z^2}{2}\Big)Y_n, \qquad R(z) = 1 + z + \frac{z^2}{2}.$$

This matches e^z up to z^2 and misses $\frac{z^3}{6}$: an error of order h^3 per step, second order.

Four stages

RK4 does the same for Simpson's rule, which needs the midpoint and the end. It predicts the midpoint twice and the end once:

$\boldsymbol{k}_1 = \boldsymbol{f}(t_n,\ \boldsymbol{Y}_n)$	slope at the start,
$\boldsymbol{k}_2 = \boldsymbol{f}(t_n + \frac{h}{2},\ \boldsymbol{Y}_n + \frac{h}{2}\,\boldsymbol{k}_1)$	slope at a midpoint predicted with $\boldsymbol{k}_1$,
$\boldsymbol{k}_3 = \boldsymbol{f}(t_n + \frac{h}{2},\ \boldsymbol{Y}_n + \frac{h}{2}\,\boldsymbol{k}_2)$	slope at a midpoint predicted with $\boldsymbol{k}_2$,
$\boldsymbol{k}_4 = \boldsymbol{f}(t_n + h,\ \boldsymbol{Y}_n + h\,\boldsymbol{k}_3)$	slope at an end point predicted with $\boldsymbol{k}_3$.

Then it applies Simpson's rule with the midpoint slope taken as the average of $\boldsymbol{k}_2$ and $\boldsymbol{k}_3$:

$$\boldsymbol{Y}_{n+1} = \boldsymbol{Y}_n + \frac{h}{6}\Big(\boldsymbol{k}_1 + 4\cdot\frac{\boldsymbol{k}_2+\boldsymbol{k}_3}{2} + \boldsymbol{k}_4\Big) = \boldsymbol{Y}_n + \frac{h}{6}(\boldsymbol{k}_1 + 2\boldsymbol{k}_2 + 2\boldsymbol{k}_3 + \boldsymbol{k}_4).$$

The method is explicit: four evaluations of $\boldsymbol{f}$ per step and no solves. If $\boldsymbol{f}$ depends on t only, then $\boldsymbol{k}_2 = \boldsymbol{k}_3$ and the update is Simpson's rule itself. Why two midpoint slopes rather than one? The test equation answers this.

The check on the test equation

For $\dot{y} = \mu y$ every trial point is Y_n times a polynomial in z, and every slope is μ times the trial point:

$$\begin{aligned} k_1 &= \mu Y_n, \\ Y_n + \tfrac{h}{2}k_1 &= (1+\tfrac{z}{2})Y_n, & k_2 &= \mu(1+\tfrac{z}{2})Y_n, \\ Y_n + \tfrac{h}{2}k_2 &= (1+\tfrac{z}{2}+\tfrac{z^2}{4})Y_n, & k_3 &= \mu(1+\tfrac{z}{2}+\tfrac{z^2}{4})Y_n, \\ Y_n + hk_3 &= (1+z+\tfrac{z^2}{2}+\tfrac{z^3}{4})Y_n, & k_4 &= \mu(1+z+\tfrac{z^2}{2}+\tfrac{z^3}{4})Y_n. \end{aligned}$$

Collect the coefficients of μY_n in $k_1 + 2k_2 + 2k_3 + k_4$:

	1	z	z^2	z^3
k_1	1	0	0	0
$2k_2$	2	1	0	0
$2k_3$	2	1	$\frac{1}{2}$	0
k_4	1	1	$\frac{1}{2}$	$\frac{1}{4}$
sum	6	3	1	$\frac{1}{4}$

So $k_1 + 2k_2 + 2k_3 + k_4 = \mu(6 + 3z + z^2 + \frac{z^3}{4})Y_n$, and with $\frac{h}{6}\,\mu = \frac{z}{6}$,

$$Y_{n+1} = Y_n + \frac{z}{6}\Big(6 + 3z + z^2 + \frac{z^3}{4}\Big)Y_n = \Big(1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24}\Big)Y_n.$$

The factor $R(z) = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24}$ is e^z up to z^4 and misses $\frac{z^5}{120}$: one step is in error by $\mathcal{O}(h^5)$, the error at a fixed time is $\mathcal{O}(h^4)$, and halving h divides it by about 16.

The second midpoint slope is what makes the z^3 term right. With $4k_2$ in place of $2k_2 + 2k_3$ the sum would be $6 + 3z + \frac{z^2}{2} + \frac{z^3}{4}$, and $R(z) = 1 + z + \frac{z^2}{2} + \frac{z^3}{12} + \frac{z^4}{24}$, with $\frac{1}{12}$ where e^z has $\frac{1}{6}$: only second order.

For a system $\dot{\boldsymbol{y}} = \boldsymbol{A}\boldsymbol{y}$ the same algebra holds with μ replaced by $\boldsymbol{A}$, since powers of $\boldsymbol{A}$ commute. The growth matrix is the same polynomial,

$$\boldsymbol{G}_{RK} = \boldsymbol{I} + h\boldsymbol{A} + \frac{(h\boldsymbol{A})^2}{2} + \frac{(h\boldsymbol{A})^3}{6} + \frac{(h\boldsymbol{A})^4}{24},$$

the Taylor series of $e^{h\boldsymbol{A}}$ up to the fourth power.

Fourth order for any f: the Taylor check

We now check, with every step written out, that one RK4 step agrees with the Taylor series of the true solution up to the h^4 term. To keep the bookkeeping short take a single equation that does

not depend on t explicitly, $\dot y = f(y)$, start from $Y_n = y(t_n)$, and write f, f', f'', f''' for f and its derivatives with respect to y, all evaluated at Y_n.

The true solution. Differentiate $\dot y = f(y)$ with the chain rule, using $\dot y = f$ each time:

$$\begin{aligned}
\ddot y &= f'\,\dot y = f f',\\
\dddot y &= \frac{d}{dt}(f f') = (f'\,f' + f\,f'')\dot y = f f'^2 + f^2 f'',\\
y^{(4)} &= \frac{d}{dt}(f f'^2 + f^2 f'') = (f'^3 + 2f f' f'' + 2f f' f'' + f^2 f''')\dot y = f f'^3 + 4 f^2 f' f'' + f^3 f'''.
\end{aligned}$$

So the Taylor series of the true solution over one step is

$$y(t_{n+1}) = Y_n + hf + \frac{h^2}{2} f f' + \frac{h^3}{6}(f f'^2 + f^2 f'') + \frac{h^4}{24}(f f'^3 + 4 f^2 f' f'' + f^3 f''') + \mathcal{O}(h^5).$$

The stages. Each stage is f at a shifted point, $f(Y_n + \delta)$, and Taylor's theorem in δ gives

$$f(Y_n + \delta) = f + \delta\, f' + \frac{\delta^2}{2} f'' + \frac{\delta^3}{6} f''' + \cdots.$$

We keep every term up to h^3 in each stage (multiplied by h in the update, that is h^4).

- $k_1 = f$.
- k_2: here $\delta = \frac{h}{2} k_1 = \frac{h}{2} f$, so
$$k_2 = f + \frac{h}{2} f f' + \frac{h^2}{8} f^2 f'' + \frac{h^3}{48} f^3 f''' + \mathcal{O}(h^4).$$
- k_3: here $\delta = \frac{h}{2} k_2 = \frac{h}{2} f + \frac{h^2}{4} f f' + \frac{h^3}{16} f^2 f'' + \cdots$, with $\delta^2 = \frac{h^2}{4} f^2 + \frac{h^3}{4} f^2 f' + \cdots$ and $\delta^3 = \frac{h^3}{8} f^3 + \cdots$. Then
$$\begin{aligned}
\delta\, f' &= \frac{h}{2} f f' + \frac{h^2}{4} f f'^2 + \frac{h^3}{16} f^2 f' f'' + \cdots,\\
\frac{\delta^2}{2} f'' &= \frac{h^2}{8} f^2 f'' + \frac{h^3}{8} f^2 f' f'' + \cdots,\\
\frac{\delta^3}{6} f''' &= \frac{h^3}{48} f^3 f''' + \cdots,
\end{aligned}$$
and adding up ($\frac{1}{16} + \frac{1}{8} = \frac{3}{16}$),
$$k_3 = f + \frac{h}{2} f f' + h^2\Big(\frac{1}{4} f f'^2 + \frac{1}{8} f^2 f''\Big) + h^3\Big(\frac{3}{16} f^2 f' f'' + \frac{1}{48} f^3 f'''\Big) + \mathcal{O}(h^4).$$
- k_4: here $\delta = h k_3 = hf + \frac{h^2}{2} f f' + h^3(\frac{1}{4} f f'^2 + \frac{1}{8} f^2 f'') + \cdots$, with $\delta^2 = h^2 f^2 + h^3 f^2 f' + \cdots$ and $\delta^3 = h^3 f^3 + \cdots$. Then
$$\begin{aligned}
\delta\, f' &= h\, f f' + \frac{h^2}{2} f f'^2 + h^3\Big(\frac{1}{4} f f'^3 + \frac{1}{8} f^2 f' f''\Big) + \cdots,\\
\frac{\delta^2}{2} f'' &= \frac{h^2}{2} f^2 f'' + \frac{h^3}{2} f^2 f' f'' + \cdots,\\
\frac{\delta^3}{6} f''' &= \frac{h^3}{6} f^3 f''' + \cdots,
\end{aligned}$$
and adding up ($\frac{1}{8} + \frac{1}{2} = \frac{5}{8}$),
$$k_4 = f + h\, f f' + h^2\Big(\frac{1}{2} f f'^2 + \frac{1}{2} f^2 f''\Big) + h^3\Big(\frac{1}{4} f f'^3 + \frac{5}{8} f^2 f' f'' + \frac{1}{6} f^3 f'''\Big) + \mathcal{O}(h^4).$$

The weighted sum. Collect the coefficients of $k_1 + 2k_2 + 2k_3 + k_4$, term by term:

	f	$h\,ff'$	$h^2ff'^2$	h^2f^2f''	$h^3ff'^3$	$h^3f^2f'f''$	h^3f^3f'''
k_1	1	0	0	0	0	0	0
$2k_2$	2	1	0	$\frac{1}{4}$	0	0	$\frac{1}{24}$
$2k_3$	2	1	$\frac{1}{2}$	$\frac{1}{4}$	0	$\frac{3}{8}$	$\frac{1}{24}$
k_4	1	1	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{5}{8}$	$\frac{1}{6}$
sum	6	3	1	1	$\frac{1}{4}$	1	$\frac{1}{4}$

So

$$k_1 + 2k_2 + 2k_3 + k_4 = 6f + 3h\,ff' + h^2(ff'^2 + f^2f'') + \frac{h^3}{4}(ff'^3 + 4f^2f'f'' + f^3f''') + \mathcal{O}(h^4),$$

and multiplying by $h/6$,

$$Y_{n+1} = Y_n + hf + \frac{h^2}{2}\,ff' + \frac{h^3}{6}(ff'^2 + f^2f'') + \frac{h^4}{24}(ff'^3 + 4f^2f'f'' + f^3f''') + \mathcal{O}(h^5).$$

This is the Taylor series of the true solution, term by term up to h^4. One step makes an error of order h^5, and over a fixed time the error is of order h^4: RK4 is **fourth order**. The half steps and the weights $1, 2, 2, 1$ are what make every column of the table agree with the Taylor series (Kutta found them in 1901). The third stage is needed for this: with $4k_2$ in place of $2k_2 + 2k_3$, the $h^2ff'^2$ column would sum to $\frac{1}{2}$ instead of 1. For a system the same calculation has more terms, since f' becomes the Jacobian matrix and f'' a bilinear form, and RK4 matches them as well.

Example S.1 (One RK4 step for $\dot{y} = y$)**. Given.** $\dot{y} = y$, $y(0) = 1$, one step $h = 0.5$.

Find. What are $k_1, \ldots, k_4$ and Y_1? How does Y_1 compare with $e^{0.5}$, and with Simpson's rule for the same step (Example 2.1)?

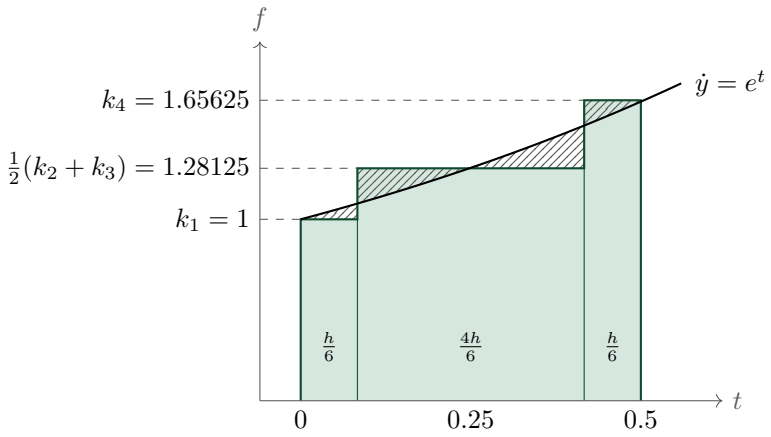
Answer. Each slope is the trial value itself, since $f(y) = y$:

$$\begin{aligned} &k_1 = 1, \\ &Y_0 + 0.25\,k_1 = 1.25, \qquad k_2 = 1.25, \\ &Y_0 + 0.25\,k_2 = 1.3125, \quad k_3 = 1.3125, \\ &Y_0 + 0.5\,k_3 = 1.65625, \quad k_4 = 1.65625. \end{aligned}$$

Then $k_1 + 2k_2 + 2k_3 + k_4 = 1 + 2.5 + 2.625 + 1.65625 = 7.78125$ and

$$Y_1 = 1 + \frac{0.5}{6} \cdot 7.78125 = 1.6484375,$$

which is also $R(0.5) = 1 + 0.5 + 0.125 + 0.0208333 + 0.0026042$. The exact value is $e^{0.5} = 1.6487213$, so the error is 0.00028, close to the first missing term $\frac{z^5}{120} = 0.00026$. Simpson's rule with the true values 1, $e^{0.25}$ and $e^{0.5}$ gives 1.6487352, an error of -0.000014. RK4 has to predict those values: its midpoint slope is $\frac{1}{2}(k_2 + k_3) = 1.28125$ against $e^{0.25} = 1.28403$, and its end slope 1.65625 against 1.64872. Most of its error comes from these predictions. The sketch shows the step as an area.



One RK4 step as an area, for $\dot{y} = y$, $y(0) = 1$, $h = 0.5$, drawn to scale. Each slope is the height of a rectangle whose width is its weight: $k_1 = 1$ over $h/6$, the average of $k_2 = 1.25$ and $k_3 = 1.3125$ over $4h/6$, and $k_4 = 1.65625$ over $h/6$. The three rectangles have the area $\frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4) = 0.6484375$, the same number as Simpson's area under the parabola through the three heights; the area under the curve $\dot{y} = e^t$ is $e^{0.5} - 1 = 0.6487213$. Where the rectangles stand above the curve and where they fall below it (hatched), the pieces almost cancel.

Runge–Kutta in practice

Library solvers such as MATLAB's `ode45` and SciPy's `solve_ivp` (by default) use a pair of Runge–Kutta formulas of orders 4 and 5, due to Dormand and Prince, that share their evaluations of $\boldsymbol{f}$. The difference between the two results estimates the error of the step, and the solver adjusts h to keep that estimate below a tolerance.